

Digital Transformation

Class # 6: Beyond Technology - Design principles for digital business

EPITA | Fall 2025

Valeriu Petruian

Digital Transformation Class 6



- Admin
- Recap
- Thoughts for the day
- Preparation for final presentations

Preparation for final presentations

- **Today (Class 6) we shall create the groups for the final sessions**
 - 8 minutes per student for the final presentation
 - DSA, AIS, SE– 2 sessions per group | CS – 1 session
 - Students presenting together – in the same session
- **QUIZ – CLOSED-BOOK EXAM**
 - Quiz is scheduled on January 15th from 09:00 to 09:30 am, in KB003 (amphi 3)
 - Students who haven't joined to date the Teams group created for the course will not be allowed to take the Quiz.

Final Presentation Sessions - Fall 25

DSA		AIS		SE		CS
S1 - 24	S2 - 20	S1 - 21	S2 - 18	S1 - 18	S2 - 18	S1 - 20

Mon Jan 19 th		Tue Jan 20 th		Wed Jan 21 st		Thu Jan 22 nd	
DSA – 7 h				[Session 1] 09:00 – 13:00			
				[Session 2] 14:00 – 17:00			
AIS – 7 h						[Session 1] 09:00 – 13:00	
						[Session 2] 14:00 – 17:00	
SE – 6 h	[Session 1] 09:00 – 12:00	[Session 2] 13:00 – 16:00					
CS – 3 hours	[Session 1] 13:00 – 16:00						

Course Breakdown

EPITA Fall 2025

Class	Date & Time Topics
Class # 1	Welcome to the Digital Economy!
Class # 2	Technology Drivers # 1: Internet of Things (IoT), Blockchain
Class # 3	Technology Drivers # 2: Big Data & Artificial Intelligence (AI)
Class # 4	Social (non-technical) drivers of Digital Transformation
Class # 5	Digital Business Models - Platforms
Class # 6	Beyond Technology - Design principles for digital business Course Wrap-Up
Class # 7	Final Presentations

Today's Reading:

- **Jeannette M. Wing.** *Computational thinking and thinking about computing*. Phil. Trans. R. Soc. A (2008) 366, 3717–3725 doi:10.1098/rsta.2008.0118



Class # 6

Beyond technology: Design principles for digital business

Economic properties of digital goods and digital products

Beyond technology: value creation in the digital economy

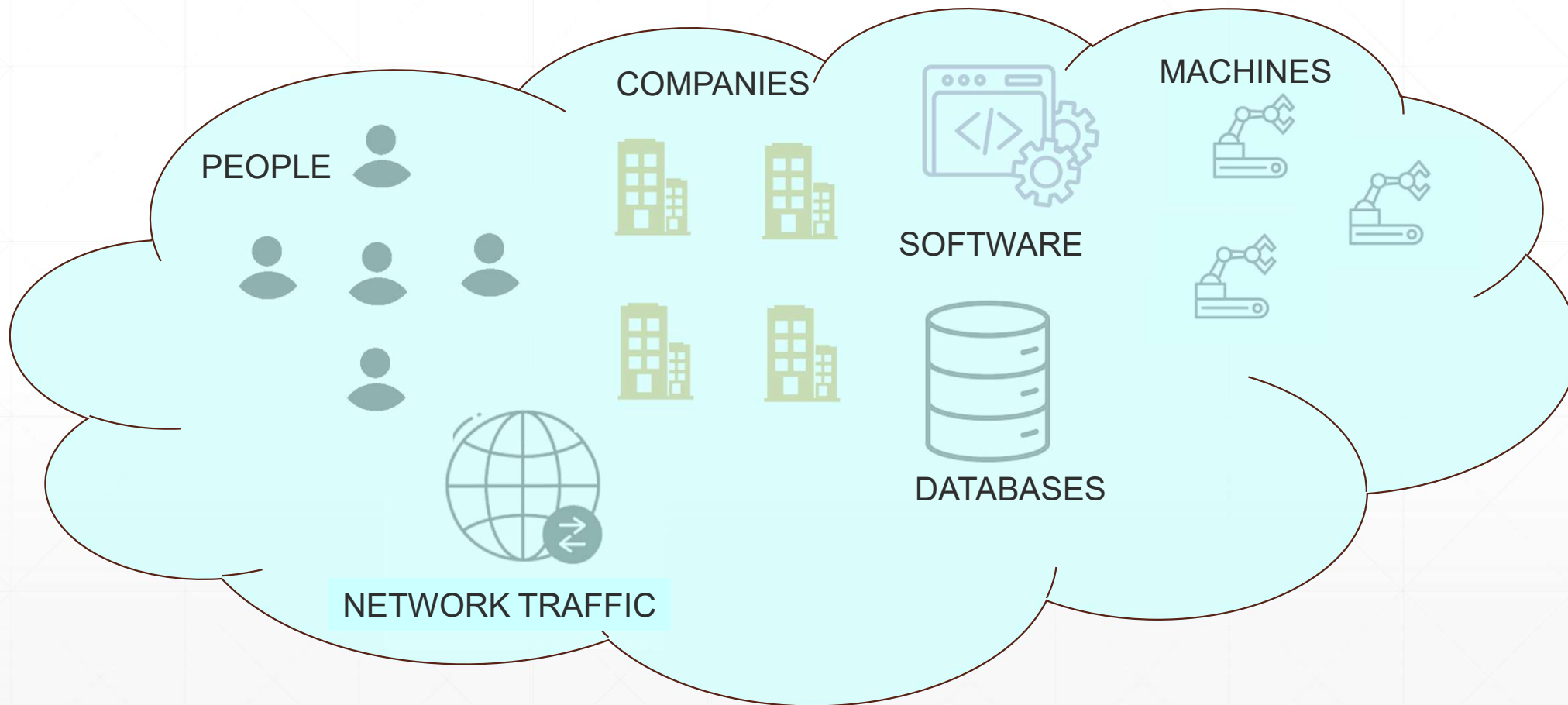
Designing digital businesses and markets

Design principles for Digital Business

Preliminary considerations

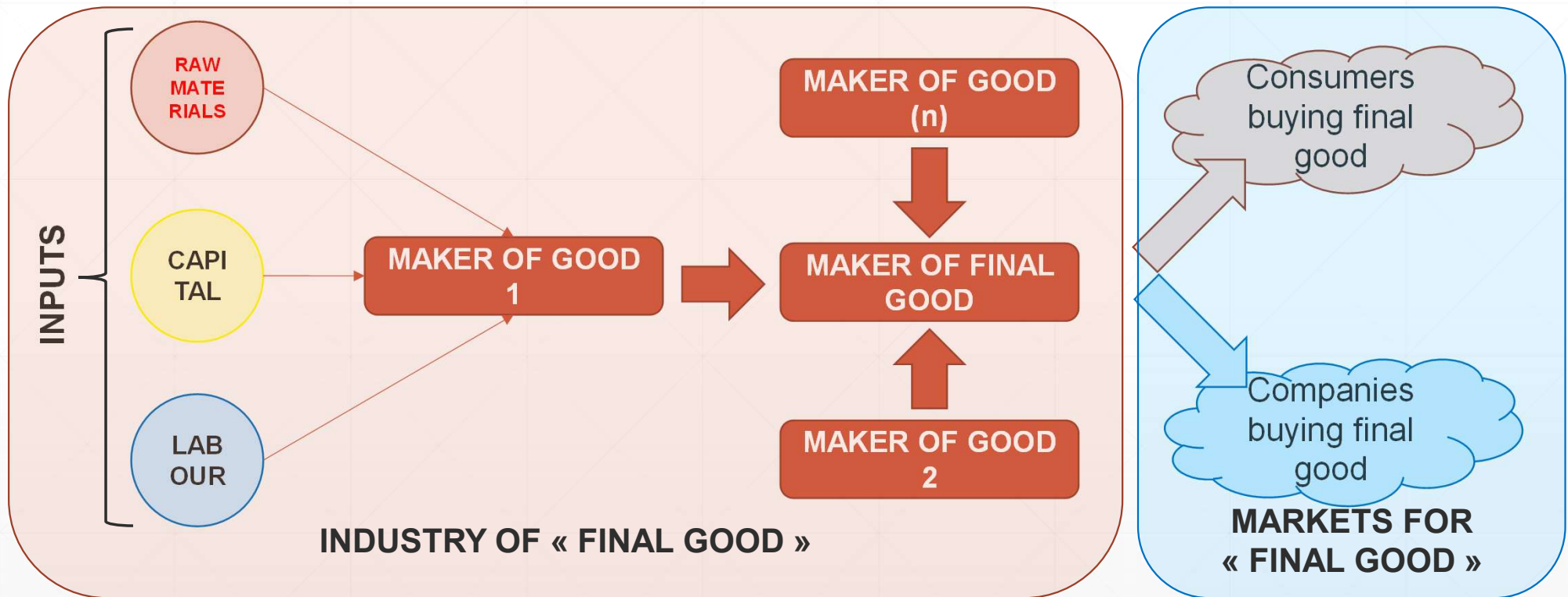
- In the previous classes we have talked:
 - About the relationship between technology adoption and diffusion and the emergence of dominant designs (Class 1)
 - About the way several digital technologies are utilized for business purposes (Classes 2 and 3)
 - About the necessity to redesign some digital products to avoid distortions and misappropriation (Class 4)
 - About the ingredients of platforms as business models (Class 5)
- Today's class provides a unified perspective on the various concepts, mechanisms, and tools that may be mobilized when designing digital businesses

The Internet = People + Companies + Machines + Software + Data + Network Traffic



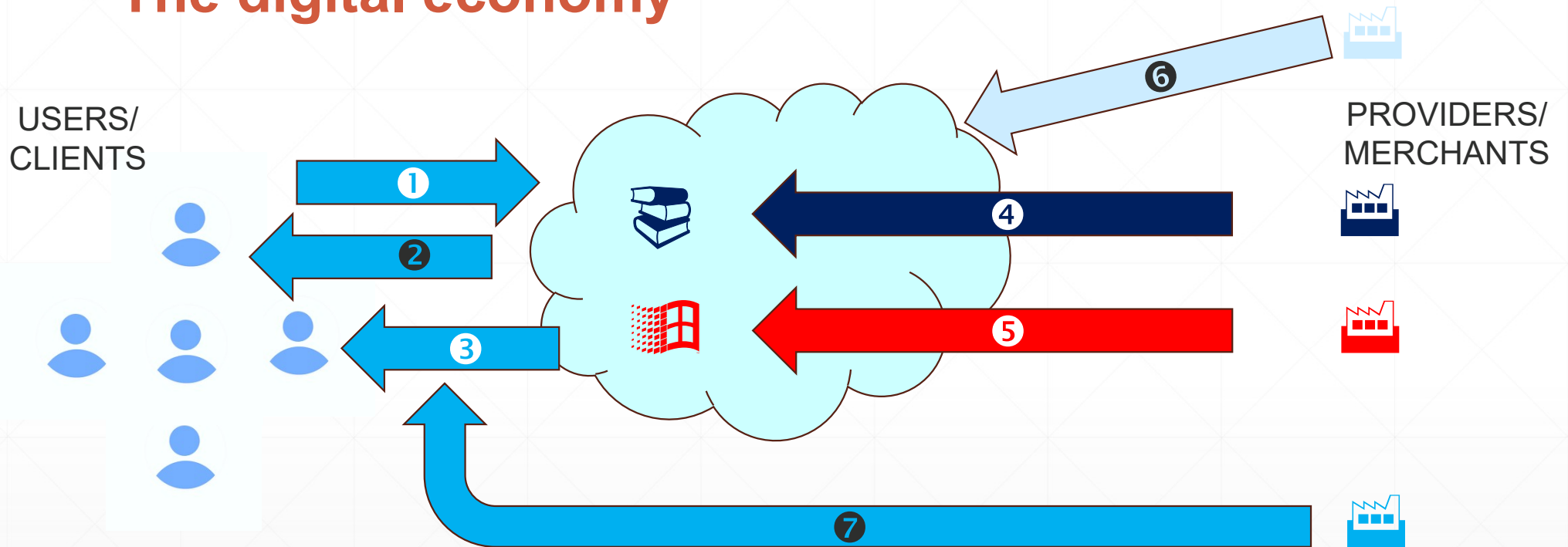
The traditional economy

Reminder



NB. In a market economy, the « inputs, » good 1, ..., good (n), are tradeable as well, hence « markets. »

The digital economy



In the Digital Economy, in addition to the traditional economic institutions (industry and markets) several other specific mechanisms need to happen

Designing Digital Businesses

Class 6

- Digital Goods (Products & Services) – Overview, main economic properties
- Digital Markets – Overview
- Transformation of existing Business Models
- Digital Transformation of organizations

Digital Goods = Digital products, digital services

PRODUCTS

- Virtual
- **Zero cost** to producing extra unit (marginal cost)
- Can be shared over the Internet (i.e. network good)
- Has some form of value (financial, psychological, usage, ...) for its users

SERVICES

- « Invisible »
- Rely on **enabling technologies** such as network access and transmission of data over fixed and mobile networks, as well as data storage and data processing
- For convenience, we group all these into **cloud computing**, also called Anything as a Service -XaaS

Physical (Tangible) and Digital Goods Comparison

	Type of good	Type of usage	Marginal cost	Exchange
Tangible good	Rival	Stock	Non-negligible	Exclusivity
Digital good	Non-Rival	Flow	Zero	Duplication



What is a digital product?

Examples

DIGITAL

- Microsoft Office documents (.docx, .pptx, .xlsx)
- Webpages on the Internet
- Song tracks on Spotify
- Applications on iOS or Android app stores
- Data within electronic bank accounts
- Apartment listing on Airbnb
- Profiles in social networks

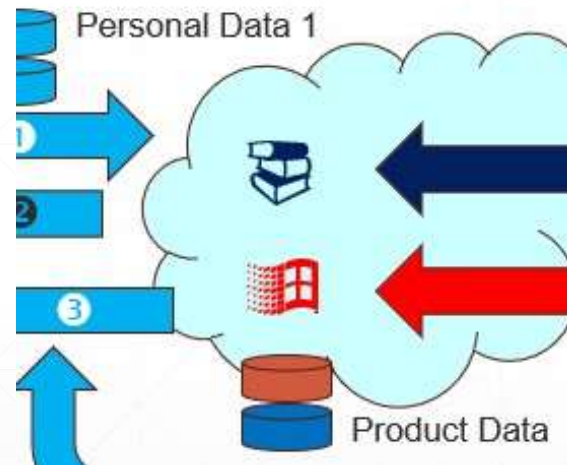
NON-DIGITAL

- iPhone, Galaxy phone
- Laptop computers
- Switches, routers, radio base stations, set-top boxes
- Sensors embedded in objects for IoT connectivity
- SIM cards

Digital Good, Tangible Good

DIGITAL GOOD

- NON-RIVAL
- REPLICABLE AT ZERO ADDITIONAL COST
- NEEDS ADDITIONAL GOODS/SERVICES FOR CONSUMPTION

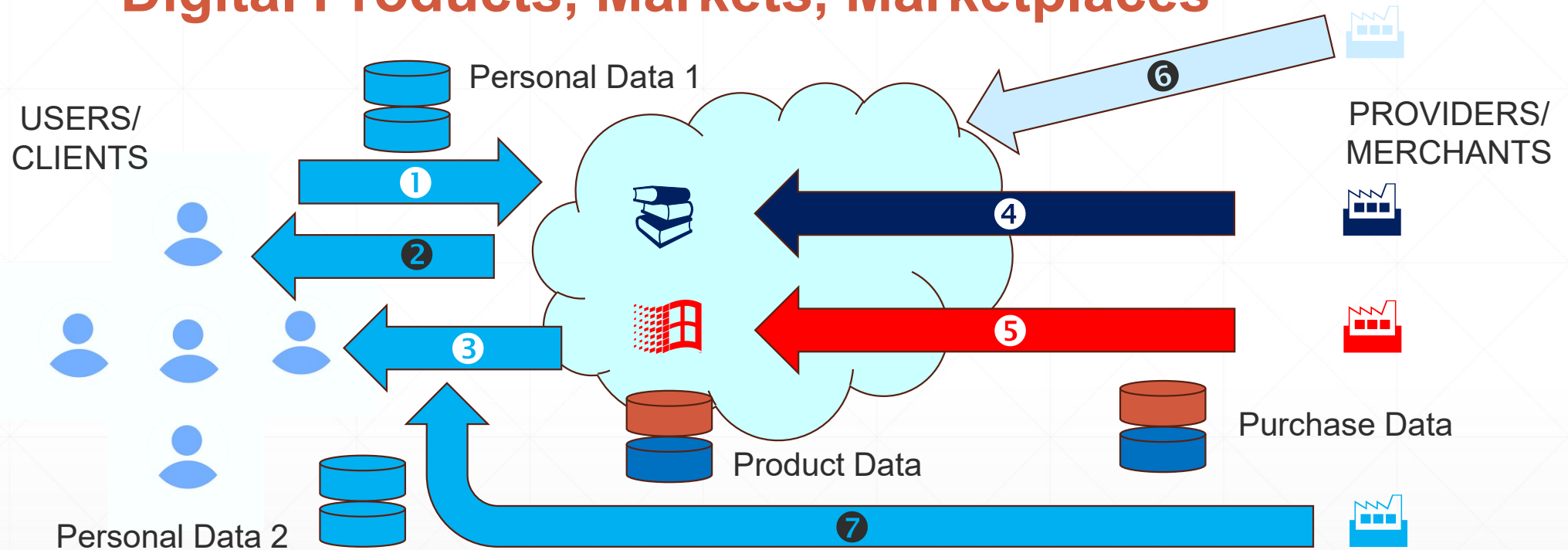


TANGIBLE GOOD

- RIVAL
- NON NEGLIGIBLE MARGINAL COST
- CAN BE CONSUMED AS SUCH

Important distinction: “*Search good* is a good with attributes or qualities that can be evaluated before consumption. *Experience good* is good with attributes and qualities that can only be evaluated after consumption. » Source : (Øverby, Audestad 2018)

Digital Products, Markets, Marketplaces



- ❶ Accesses online marketplace
- ❷ Buys tangible good online
- ❸ Buys digital good online

- ❹ Sells tangible good online
- ❺ Sells digital good online

- ❻ Provides infrastructure for online marketplace
- ❼ Provides access to online marketplace

Digital Products, Markets, Marketplaces

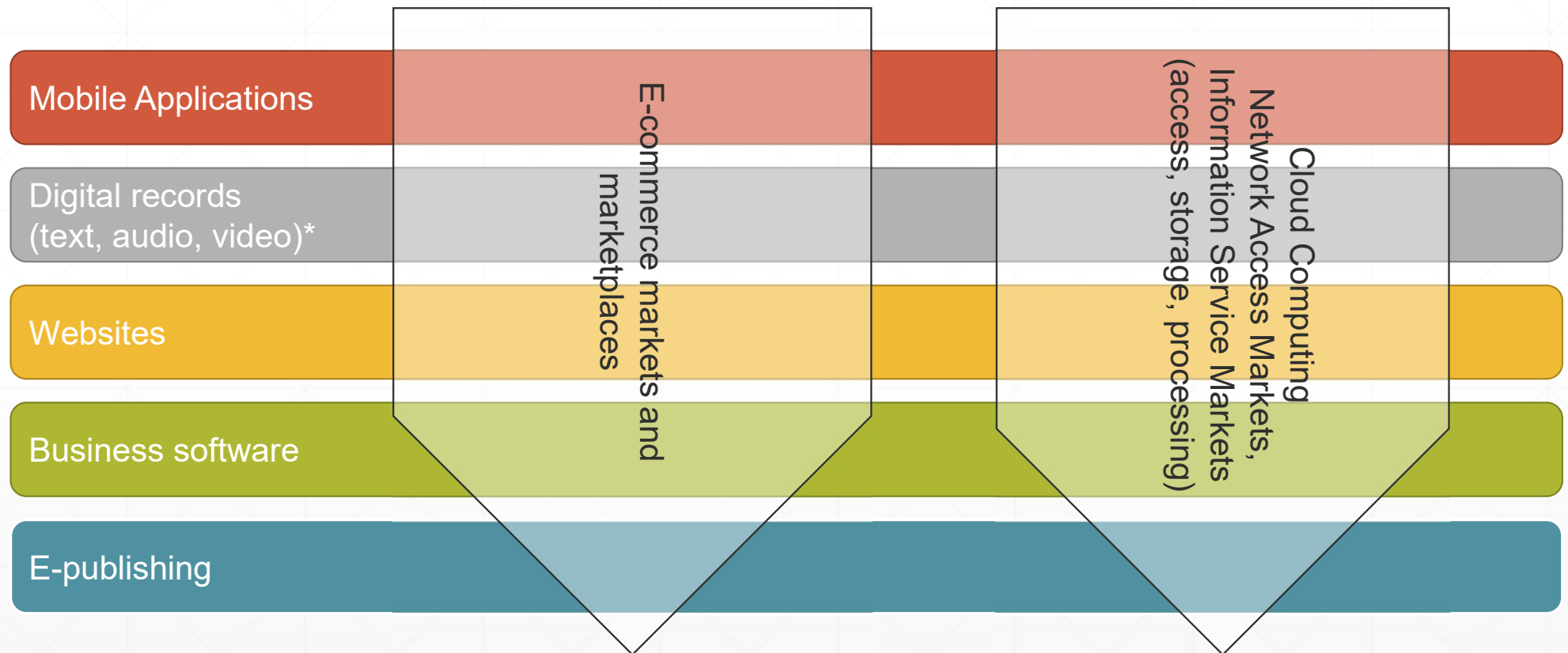
Digitalisation of goods, transactions, exchange mechanisms

PRODUCT	MARKET	MARKETPLACE
Tangible good: rival, marginal cost non negligible	For tangible goods: transaction	Built-in mechanism for displaying products: catalogue
Digital good: non-rival, zero marginal cost	For digital goods: transaction and/or subscription	Built-in mechanism for price setting: ex. auctions, dynamic pricing
	For access to online: access fees	Built-in mechanism for matching: ex. recommendation
	For usage of online services: usage fees,	Built-in mechanism for dispute resolution: customer service

Ex: any company selling online

Ex: Amazon, ebay

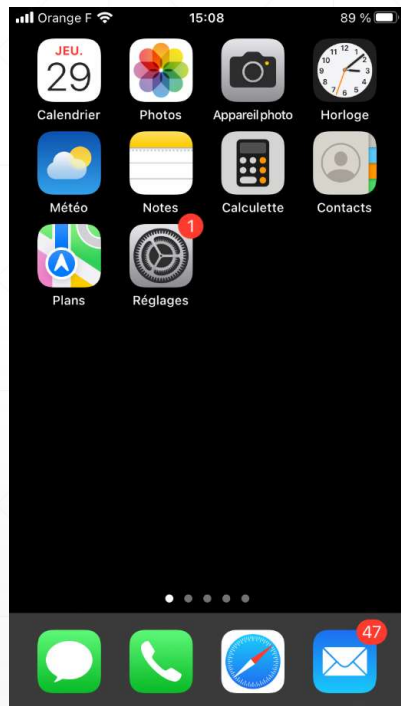
Digital Products, Digital Markets - Overview



* AirBnB apartment listing, song track on Spotify, movie track on Netflix

Digital Products and Services

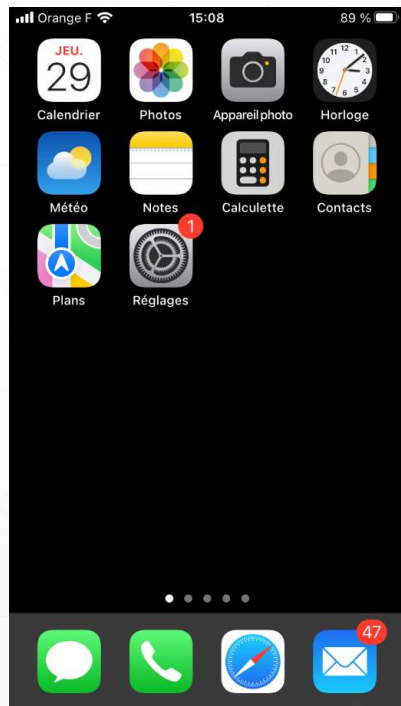
Mobile Applications 1/2



- Economies of scale
 - Once we have produced it, any additional (N+1) copy comes at zero marginal cost
- Development environment (SDK, API)
- Costs:
 - Development
- Revenues
 - Per unit (transaction)
 - Advertising
 - Subscription

Digital Products and Services

Mobile Applications 2/2



- Channels
 - App Stores
- Relationships
 - 1-2-1, 1-2-many
- Partners
 - Depending on BM

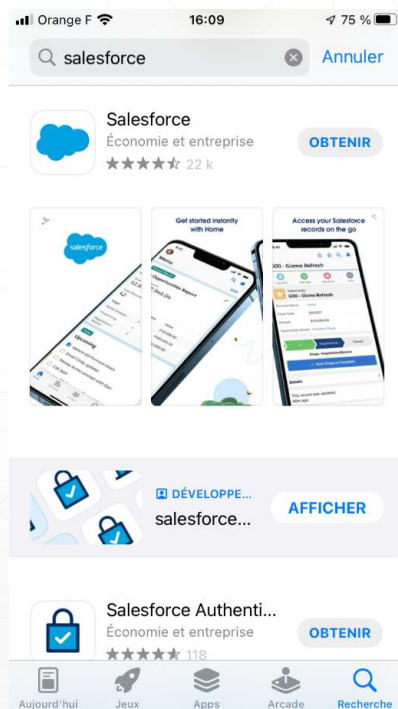
Digital Products & Services Websites



- Channel-type of interactional « product »
- Developed as a product (outside party) or as a digital transformational initiative (inside party)
- « Corporate » dependencies
- In recent years, their combination with « mobile apps » has drastically reduced their impact
- Nonetheless, they remain a « prerequisite » in all entrepreneurial ventures

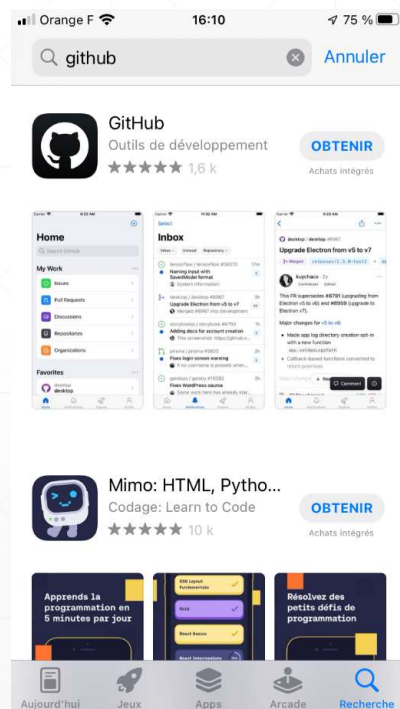
Digital Products & Services

Business Software (proprietary)



- Subscription (or license) based
- Develop once, publish many (in the cloud)
- Compatibility
 - Operating systems
 - Ecosystem
 - Existing legacy environment

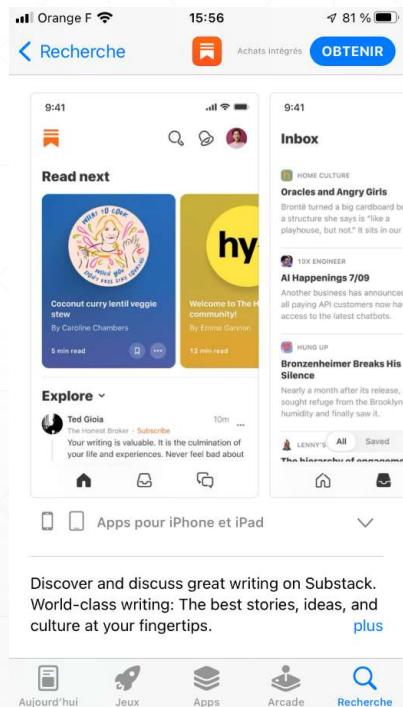
Digital Products & Services Business Software (open source)



- Collaborative type of approach
- Innovation platform
- Free to use, modify
- Shared IP
- Participatory entrepreneurship

Digital Products & Services

e-publishing



- Newsletters, e-books, ...
- Standards
 - Open: .pdf, .html, ...
 - Proprietary: Amazon Kindle
- 1 – 2 – many
- Revenue models:
 - Subscription
 - Ad-generated revenues

Digital markets

Main players, Main types

MAIN PLAYERS

- Network Provider
- Internet Service Provider
- Application Service Provider
- Content Provider

In recent years:

- NP and ISP roles tend to merge
- ASP and CP roles as well

MAIN TYPES

- e-commerce markets
- Cloud computing - Network access markets, Information services markets (access, storage, processing)

Digital markets

e-commerce

- e-commerce (including m-commerce) markets
 - Selling/buying on-line all types of products and services (digital or tangible)
 - Digital products and sold and delivered online
 - Tangible products are sold online and delivered physically (transport)
- What is digital?
 - « on-line » or « e » - means virtual purchase experience

NB. What distinguishes e-commerce from other online activities is that there is payment involved

Digital markets

Cloud computing (access, storage, processing)

- The “invisible engines” of the digital economy: silent, but ubiquitous
- Types:
 - Private
 - Public
 - Hybrid
- Illustrative examples:
 - Netflix and AWS
 - Spotify and Google Cloud
- Intangible but not immaterial (ex: data centers and greenhouse gas emissions)

Digital products, digital markets

Summary

- The economic properties of digital goods (network good, zero marginal cost) makes them a quite distinctive class of goods
- The economic value of digital goods must be considered in combination with the markets that allow their production and/or exchange
 - Example 1: a music track on Spotify has value because it can be accessed, exchanged and consumed through the applications and information services allowed by Internet access and audio streaming
 - Example 2: an entry in Wikipedia has value because it is produced and consumed through peer-production and access to a common and shared repository of content creation

What is there to be transformed in an organization?

Class Discussion



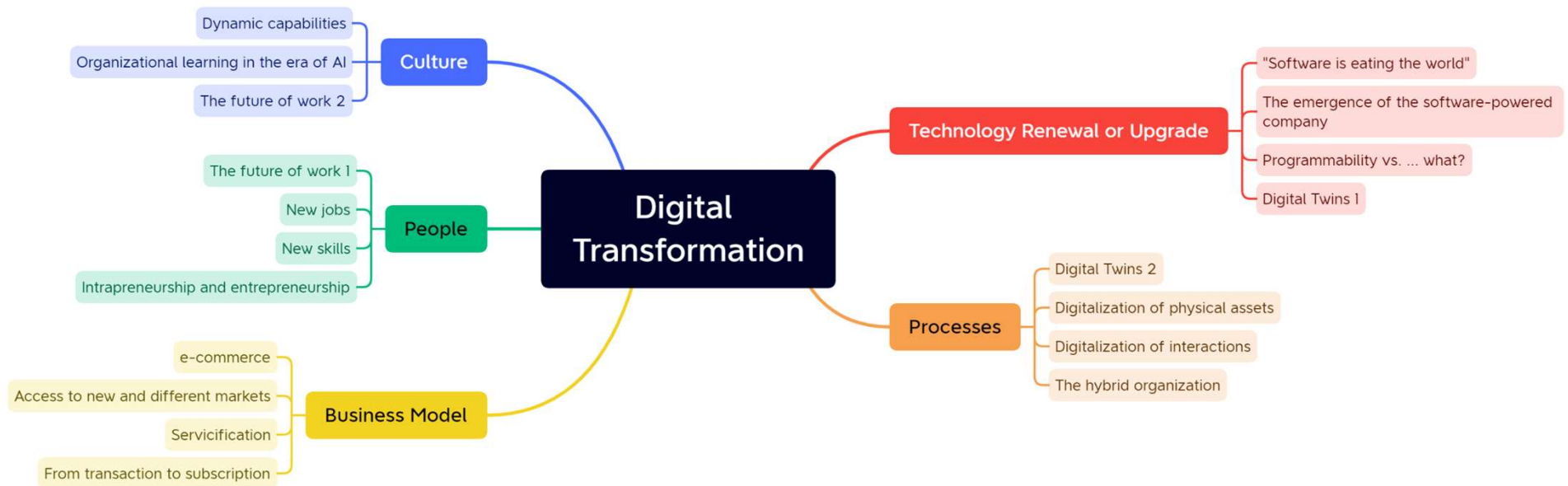
- Technology – renewal or upgrade
- Processes – organizational interactions
- Business model – the way the company operates to create value
- People – new skills, new jobs
- Culture – shared values, norms, and ways of doing things
- All of the above
- Other?

Digital Transformation of Organizations

Class 6 Wrap-Up

- Ultimately, it is at an organization level that one can observe the most intimate interplay of the forces at work in the digital economy and the network society
- It is there, at a micro level, that change is being produced, in the place where people, processes, and technology interact. It is there, at a concrete level, where the immediate consequences of “change” can be observed
- It is there, as well, where the individual and collective aspects of digital transformation can be appraised in their finest details, and where harmonious, or conflictual, interdependencies of the various forces at work can be detected

Digital Transformation of Organizations



Presented with xmind

Digital Transformation Technology

- As we have seen, much of Digital Transformation is about technology change
- Change at this level happens either as **renewal** or as **upgrade**. The first is about replacing existing technology, the second is about incrementally changing versions of the same breed of technology
 - Renewal: 5G replacing 4G
 - Upgrade: iOS version 15 replacing iOS version 14
- Either way, change is being produced. At a macro level, some forms of change are continuous (as in sustaining innovation), others may be disruptive
 - Replace physical keypad with tactile screen
 - Replace feature phone with smartphone



Is... « software eating the world? »

Class Discussion

- A vision promoted by venture capitalists and digital entrepreneurs, as expressed in Marc Andreessen's WSJ article from 2011
- Explains, to a great extent, the entrepreneurial wave originating from the Silicon Valley, that has contributed to the emergence of leading companies in various industries
- Explains also the “disruptive” nature of the emergence of such companies, who, through their incursions in established industries, have brought about change, at a macro level

BUT, IS IT TRUE THAT « SOFTWARE IS EATING THE WORLD? » COMMENTS, PLEASE?

Source: « *Why software is eating the world.* » Marc Andreessen (2011)

Virtualization, Programmability, Automation

VIRTUALIZATION

- “In computing, **virtualization** or **virtualisation** (...) is the act of creating a virtual (rather than actual) version of something at the same abstraction level, including virtual computer hardware platforms, storage devices, and computer network resources.” (Wikipedia)

PROGRAMMABILITY

- « (of a computer or other machine) able to be provided with coded instructions for the automatic performance of a task. » (Oxford Online Dictionary)

AUTOMATION

- « the use of machines and computers that can operate without needing human control.» (Cambridge Online Dictionary)

Company functions and software

Discussion



COMPANY FUNCTIONS

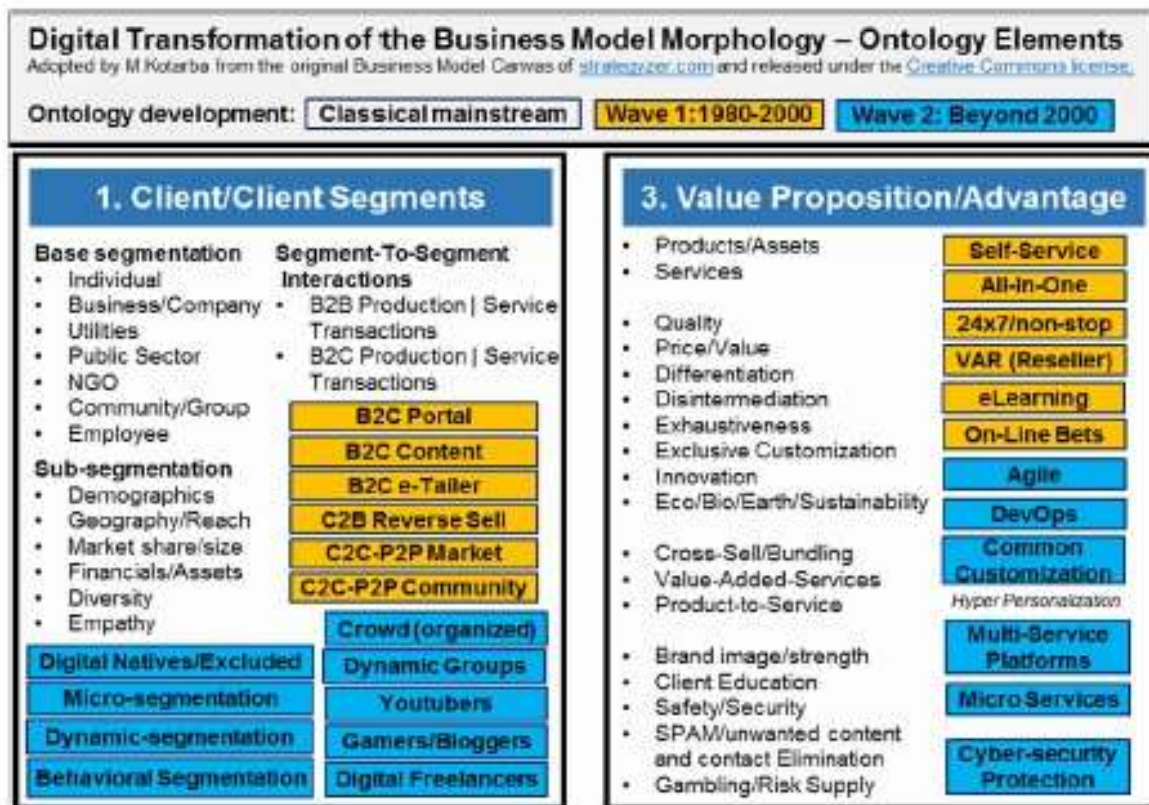
- Marketing & Sales
- Operations
- Production
- Legal
- HR
- Finance & Accounting

DEDICATED SOFTWARE

- CRM, CMS
- OMS
- SCM
- LPMS
- HCM, ESN
- ERP, RMS, B&PS, ...

The emergence of the « software-powered » company

- As software becomes more and more adopted by companies, many of them become dependent on IT and software, at a much more rapid and large scale than ever before
- Software has become, indeed, a strategic asset in most business organizations, alongside physical assets such as real estate, and capital
- This implies a redefinition of some of the company's processes, as seen with the increased deployment of new approaches and methodologies such as:
 - Agile methods
 - Lean methods
 - SOA, BPM, DEVOPS
 - User Experience (UX), Customer Experience (CEx)



Source: M. Kotarba DOI: 10.2478/fman-2018-0011

Digital Transformation of existing Business Models – Illustration (Client segments 1)

This example illustrates how businesses, while adopting newer generations of digital technologies, induce change at a Business Model component level, here: client segments

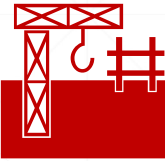
Source: M. KOTARBA, Digital Transformation of Business Models, Foundations of Management, Vol. 10 (2018), ISSN 2080-7279

2.1 Client/Client Segments – Classical Mainstream	
1. Base segmentation: a) individual, b) business/company, c) utilities, d) public sector, e) nongovernmental organizations (NGO), f) community/group.	d) financials/assets, e) diversity. 3. Empathy 4. Segment-to-segment interactions ²⁰ : a) B2B – Business to Business (standard form: production and services), b) B2B – Transaction processing (e.g., FX trading), c) B2C – Business to consumer transaction processing (e.g., home banking).
2. Sub-segmentation: a) demographics, b) geography, territorial reach, c) market share/size,	
2.1 Client/Client Segments – Wave 1 (1980–2000)	
New segment to segment business interactions: 1. B2C portal – vertical or horizontal data collection and presentation, 2. B2C content delivery, e.g., digital media sales, 3. B2C eTailor (e-Commerce retailer), such as a bookstore, with online sales and offline delivery, 4. C2B (consumer to business) – e.g., clients reverse selling previously purchased goods, 5. C2C-P2P (consumer to consumer or person to person) ²¹ market – e.g., local auction services, local classified ads, 6. C2C-P2P Community, e.g., alumni sites.	
2.1 Client/Client Segments – Wave 2 (Beyond 2000)	
1. New segments: a) digital natives, b) digitally excluded, c) digital freelancers – individuals whose digital competencies can be used across the world as long as they are able to access the Internet (programmers, designers, consultants), d) crowd (organized groups with particular interests, demands, or capacity), e) dynamic groups (e.g., participants of an event or people grouped for a short period of time in a geographical location), f) influencers promoters – people or groups shaping the opinions (both positive and negative) or decisions of others: • bloggers (online publishers and trend setters), • youtubers ²² – content generating individuals with commercial gratification (fees), • gamers (game players, shifting from digital to analog). 2. Microsegmentation (e.g., using Big Data to further explore the unique characteristics of subpopulations). 3. Dynamic segmentation (e.g., using online information from real-time sensors to identify specific segmentation parameters). 4. Behavioral segmentation (e.g., using lifetime data to observe trends and forecast their development).	

Digital Transformation of existing Business Models – Illustration (Client segments 2)

Change of technology applications at a component level may, in some cases, take into account recent social phenomena (ex: Gen Y, Gen Z) or technological enablements (Big data)

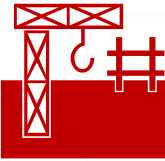
Source: M. KOTARBA, Digital Transformation of Business Models, Foundations of Management, Vol. 10 (2018), ISSN 2080-7279



Class assignment # 6

Please read the following article: **Jeannette M. Wing**. *Computational thinking and thinking about computing*. Phil. Trans. R. Soc. A (2008) 366, 3717–3725 doi:10.1098/rsta.2008.0118 (available here: <https://www.cs.cmu.edu/~wing/publications/Wing08a.pdf>). Based on today's reading and the discussion in class, please choose one of the two options below and elaborate upon it:

1. Leveraging digital technologies to transform existing business models
2. Leveraging digital technologies to create a « fully digital » business



Class assignment # 6 - continued

1) Leveraging digital technologies to transform existing business models

You may use the Business Model Canvas from the Enterprise Essentials course, find it here:

https://en.wikipedia.org/wiki/Business_model_canvas) :

- Which digital technology have you considered?
- For which business model component?
- What other types of changes do you think the introduction of your chosen technology may bring about in the organization?

2) Leveraging digital technologies to create a « fully digital » business

You may refer to Class 5: Digital business models – platforms

- Which digital technologies have you considered to power your business?
- What social need your “fully digital” business will fulfil?
- What type of change do you expect your “fully digital” business to bring about in the industry it operates?

Digital Transformation of Organizations

Data and Digital Twins

INFORMATION TECHNOLOGY

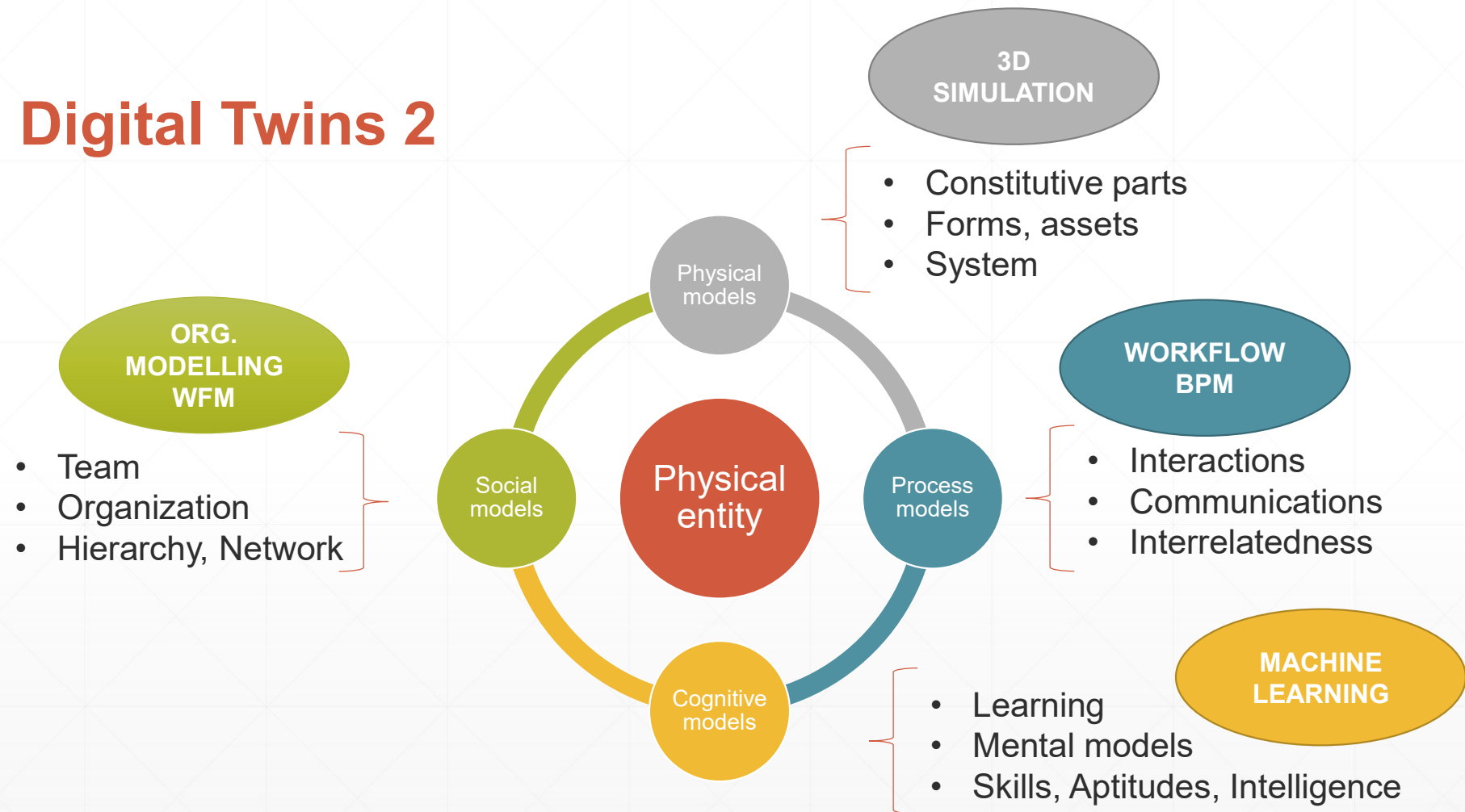
- The importance of Data and Information – for humans (decision-making), for machines (training in Machine Learning)
- « *In the 21st century, the « I » will be more important than the « T »* » (Peter Drucker)

DIGITAL TWINS

“Digital replica of a living or non-living physical entity”

- By digitalization of their **assets**, firms transform their products
- By digitalization of their **processes**, firms transform work, itself

Digital Twins 2



Digitalisation ... of physical assets

TANGIBLE GOOD « A TABLE »



DIGITAL GOOD « DATA AND INFORMATION ABOUT A TABLE »

- Who produced it? What is it made of? Dimensions, weight. Usages and functionality
- What makes this table different than other tables? (and why you should buy it)
- Information about price, where to find it, who already bought it, why they did so, and what else did the people who purchased this table bought (ideally from the same vendor).
- By the way, if you drink tea/coffee, or like reading, **this table is made just for you!** (and of course, for any additional table sold, we'll plant a tree). Incidentally, we're also making stools, are you interested?



Digitalisation ... of human interactions

HUMAN INTERACTION



DATA AND INFORMATION ABOUT HUMAN INTERACTION

- Who said what? When? Why? Who was the originator and who was the respondent?
- Did it happen by e-mail, phone, online channels, chat, instant messaging
- At which frequency? In which context?
- Whose opinion was more relevant? Whose reaction was most commented?

Image source: <https://www.geekwire.com/2018/real-human-interaction-still-preferred-texting-email-social-media-according-survey/>

Digitalisation ... of cognitive functions (e.g. decision-making)

MACHINE LEARNING

- **“Machine learning** gives computers the ability to learn without being explicitly programmed (Arthur Samuel, 1959). It is a subfield of computer science.” (Wikipedia)

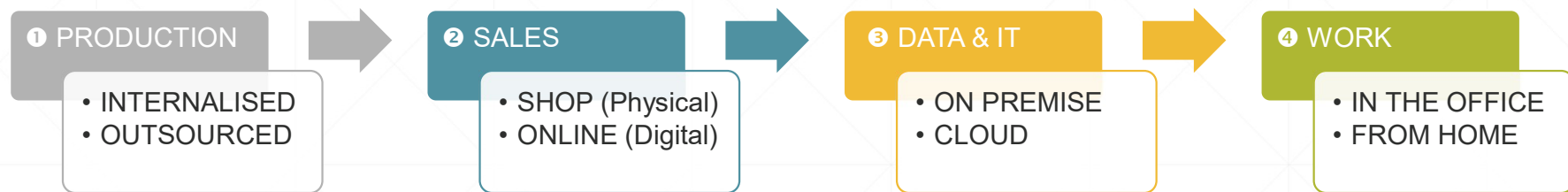
**SO, IF COMPUTERS LEARN
HOW TO BECOME FASTER,
MORE ACCURATE, MORE
EFFICIENT... WHAT SHOULD
WE, HUMANS, LEARN?**

TOP 15 SKILLS FOR 2025(*)

1. Analytical thinking and innovation
2. Active learning and learning strategies
3. Complex problem-solving
4. Critical thinking and analysis
5. Creativity, originality and initiative
6. Leadership and social influence

(*) WEF Future of Jobs Report 2020

Digitalisation In the context of the evolution of business organizations



Companies have experienced several restructurings of their fundamental operating functions, including production, sales & marketing, IT, ...

With digitalisation of human interactions and cognitive functions, they are now facing new dilemmas

Temporary trend? Pendulum shift? Time will tell, but the current debate may well contribute to ultimately redefining what a “company” is

Culture

Organizational Learning

- « Organizational learning is viewed as routine-based, history-dependent, and target-oriented. Organizations are seen as learning by encoding inferences from history into routines that guide behavior. Within this perspective on organizational learning, topics covered include how organizations learn from direct experience, how organizations learn from the experience of others, and how organizations develop conceptual frameworks or paradigms for interpreting that experience.»

Barbara Levitt, James G. March. *Organizational Learning*. Ann. Rev. Sociol. 1988

- It is by « collective learning » that we make organizations better fit to cope with their challenges, allow them to adapt and, ultimately, survive in an ever changing environment.

Culture

Collective (Organizational) Capabilities

- In organizational theory, **dynamic capability** is the capability of an organization to purposefully adapt an organization's resource base, or "the firm's ability to integrate, build, and reconfigure internal and external competences to address rapidly changing environments".

David Teece, Gary Pisano, Amy Shuen. *Dynamic Capabilities and Strategic Management*.
Strategic Management Journal, 1997

- “[successful companies] develop capabilities—an ability to innovate, for example, or to respond to changing customer needs. Such **organizational capabilities**, as we call them, are key intangible assets. You can’t see or touch them, yet they can make all the difference in the world when it comes to market value. These capabilities—the collective skills, abilities, and expertise of an organization—are the outcome of investments in staffing, training, compensation, communication, and other human resources areas. They represent the ways that people and resources are brought together to accomplish work. They form the identity and personality of the organization by defining what it is good at doing and, in the end, what it *is*. They are stable over time and more difficult for competitors to copy than capital market access, product strategy, or technology.

Norm Smallwood, Dave Ulrich. *Capitalizing on Capabilities*, Harvard Business Review, June 2004

Entrepreneurship, Intrapreneurship

ENTREPRENEURSHIP

- “**Entrepreneurship** is the creation or extraction of economic value. With this definition, entrepreneurship is viewed as change, generally entailing risk beyond what is normally encountered in starting a business, which may include other values than simply economic ones. An **entrepreneur** is an individual who creates and/or invests in one or more businesses, bearing most of the risks and enjoying most of the rewards.”

Source: Wikipedia

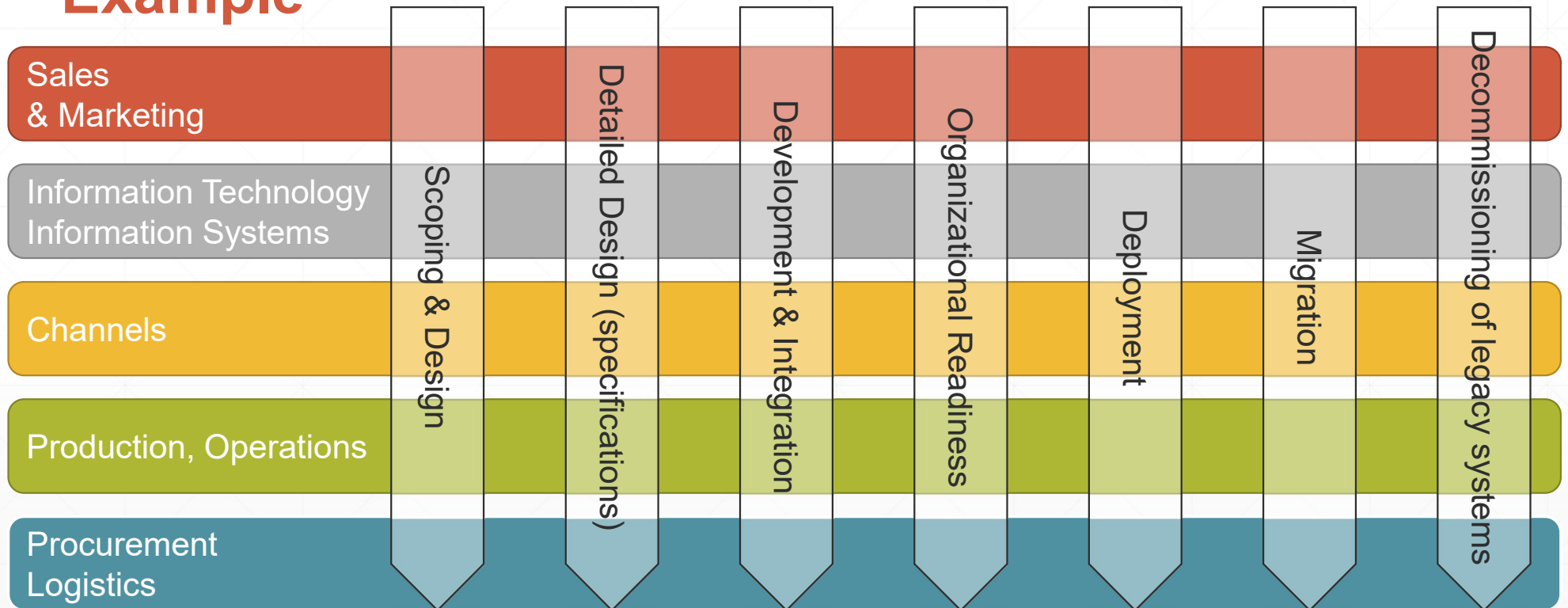
INTRAPRENEURSHIP

- “**Intrapreneurship** is the act of behaving like an entrepreneur while working within a large organization. Intrapreneurship is known as the practice of a corporate management style that integrates risk-taking and innovation approaches, as well as the reward and motivational techniques, that are more traditionally thought of as being the province of entrepreneurship.”

Digital Transformation as an Organizational Intrapreneurial Capability

- Most companies launch Digital Transformation initiatives as large, corporate-wide, endeavours, covering many aspects of the business
- It is very likely that such comprehensive initiatives are organized as **Programmes** rather than **Projects**
 - **A project** is “*a temporary organisation that is created for the purpose of delivering one or more business products according to a specified business case*”. Projects have an end and aren’t designed to last very long. The project manager ensures the project delivers the intended goal, within a defined timeframe and budget (PRINCE2)
 - **A programme** is defined as “*a group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually*”. Programmes are usually long term, sometimes spanning years, and don’t have a fixed deadline. A programme is a framework of related projects aligned in a specific sequence. They have predictable and repeatable elements to minimise or even eliminate risks. (PRINCE2)

Digital Transformation Programme Example



Digital Transformation Programme

Typical structure

Business Objective

- Why do we do it? Ex: to improve customer experience

Desired change

- What do we want to achieve from it? Ex: 50% of sales on digital channels

Transformation strategy

- How should we go about doing it? Ex: start with B2B, then B2C, first sales process (billing and payment), then ordering process

Program structure/Engagement model

- Which resources shall we mobilize? Ex: rely on an external partner or on internal resources, or a mix of both? How will the various parties engage?

Digital Transformation Programme Essentials 1

Business Process Framework

- A list of all business processes impacted by the change, both AS IS and TO BE

Architecture Blueprint

- A visual representation of all information systems, business applications and interfaces within the scope of the programme

Data Strategy

- Data Governance, Data integration, Integrated data framework, data vizualisation, data insights

Data Models

- A documented representation of the most relevant data structures (examples: customer data model, product data model)

Digital Transformation Programme Essentials 2

Programme Organization

- Main participants, main instances, timeline, risk assessment, Roles and Responsibilities, RACI matrix, budget control, dependencies matrix

Technology Change

- Technology upgrade/renewal, new applications, new interfaces

Organizational & Process Change

- Training programme, reskilling, business readiness assessment

Digital Transformation Programme Essentials 3

Executive Sponsorship

- Any digital transformation initiative should be part of the executive management agenda

Business and IT collaboration

- Even if technology is a large part of the change programme, business impacts should be carefully considered and proper involvement of all the business areas impacted by change should be carefully planned

Communication

- Effective communication should be planned and executed throughout the programme's duration

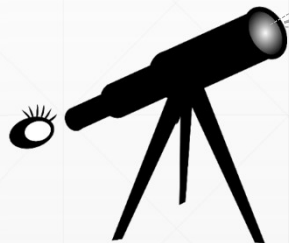
Concluding Thoughts

It is our collective responsibility to imagine a future in which we avoid “technology hijacking” and make work an ever more interesting and rewarding experience

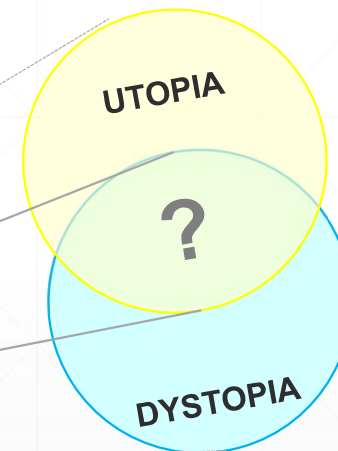


TODAY

It is up to all of us, collectively and individually, to keep technologies in their proper place, not to divert them from their initial objectives and to allow them to fulfil their role...



TOMORROW



... that of a factor of progress for the economy and the society, serving the general interest and the common good

Image source: [Cler-Free-Vector-Images](#) from [Pixabay](#)

Course Wrap-Up

What do we know about Digital Transformation?

- ❶ **Thesis # 1:** Digital Transformation is the consequence of technology adoption by individuals and organizations
- ❷ **Thesis # 2:** As technology gets adopted by a larger number of people, global infrastructures get built, supporting the entire economy and society
 - Electrical bulb, dynamo => Electricity/Energy networks
 - Telephone => Telecom networks
 - Computer => The Internet

Course Wrap-Up

What do we know about Digital Transformation ?

- ③ **Thesis # 3:** Much of the current technology adoption process is about replacing technology itself, either continuously (upgrade), or discontinuously (renewal)
- Smart phone replacing feature phone (discontinuous)
 - 5G technology replacing 4G technology
 - Cloud-based software replacing on premise IT resources
 - Big data technology replacing traditional storage and database systems
- ④ **Thesis # 4:** In the process, transformations are happening, at all levels, and in all areas:
- New jobs, new skills; technology itself creates new types of jobs that it shall destroy, as it develops itself further
 - New industries, new types of players emerge
 - Existing industries and companies get “transformed” as they adopt digital technologies

Course Wrap-Up

What do we know about digital transformation ?

- ⑤ **Thesis # 5:** Some technologies, called General Purpose Technologies, such as Artificial Intelligence, and specifically Machine Learning, have today relatively higher potential to bring about change.
- ⑥ **Thesis # 6:** Unlike other infrastructural technologies (IoT, Blockchain ...) AI and ML's potential for change encompasses activities and processes which are usually performed by humans, for example: decision-making. With AI-driven Digital Transformation, we are witnessing increased human activity embeddedness in technology.
- ⑦ **Thesis # 7:** Digital Transformation, in the era of AI, raises fundamental questions such as, for example, which activities should continue to be performed by humans, and which are the ones that could be “delegated” to AI, and in which conditions.

Course Wrap-Up

What do we know about digital transformation ?

- ⑧ **Thesis # 8:** Networks, as coordination mechanisms, are becoming central to value creation in the Digital Economy and the Network Society
- ⑨ **Thesis # 9:** Technocentric design of digital technologies leads, invariably, to the reinforcement of algorithmic agency, while sacrificing human agency
- ⑨ **“ Thesis # 9 bis:** To avoid the distortion and misappropriation caused by the forces at work in today’s digital world, we need both new forms of regulation and reimagination and redesign of digital technologies
- ⑩ **Thesis # 10:** In the Digital Economy, platforms are becoming the business model of choice. They rely on the combination of several ingredients, such as network effects, ecosystems, two-sided markets, and economies of scale.

Course Wrap-Up

What do we know about digital transformation ?

- ⑩“ **Thesis # 10 bis:** It is difficult to satisfy, over time, the needs and wants of several groups of customers affiliated to multi-sided platforms; any such loss of quality of service may result in “platform decay”
- ①① **Thesis # 11:** Digital products and markets have several distinct economic properties; therefore, the economic value of digital goods must be considered in combination with the digital markets that allow their production and/or exchange
- ①② **Thesis # 12:** It is our individual and collective responsibility to put technologies to good use, to avoid technology ‘high-jacking’, and **to allow them to fulfil their primary role: that of a factor of progress for the economy and the society, serving the general interest and the common good**

Thank You!

Valeriu Petrulian